

Critical Behaviour for Riemann zeta function? Sharp Transition

O. Shanker, Mountain View, CA 94041, U. S. A. email: oshanker@gmail.com

Abstract

We show that there is a sharp transition, reminiscent of the transitions in the theory of critical phenomenon, in the behaviour of the forward to backward symmetry in the distribution of Gram blocks for the Riemann zeta function.

Keywords: Critical behaviour, Riemann zeta, **Mathematics Subject Classification (MSC):** 11M06, 11-04.

1 Introduction

This report will be very brief. The key point is that some properties of the Riemann zeta function (related to the forward to backward symmetry of the function value distribution along the critical axis) show a sharp transition at the usual Gram points. They diverge exponentially at points before the Gram points, and decay exponentially at points after the Gram points. At the Gram points, they take on the value 1. This crossover behaviour is reminiscent of the behaviour seen in the theory of critical phenomena. The Gram blocks where we observe the phenomena are islands of non-zero $S(T)$ within the sea of zero $S(T)$.

2 Materials and Methods

We assume the reader is familiar with the usual notations for the Riemann zeta function. We present here only the minimum notation needed for our discussion. For more notation and background, see Ref [1].

2.1 Notation

In analogy with usual Gram points, we can define generalized Gram points [2] and associate an angle ϕ with a point t on the critical axis as follows:

Definition 2.1. For $t \geq 7$, t is said to be a generalized Gram point with value ϕ if $\theta(t) = 2k\pi + \phi$, where $0 \leq \phi < 2\pi$.

In what follows, when we refer to Gram point, we mean a generalized Gram point. A generalized Gram point with $\phi = 0$ corresponds to the usual Gram point with even Gram index, and a generalized Gram point with $\phi = \pi$ corresponds to the usual Gram point with odd Gram index.

Definition 2.2. A generalized Gram point g_n is called good if the real part of the Riemann zeta function is positive at the point, and bad otherwise.

Definition 2.3. A Gram block is an interval $[g_n, g_{n+k})$ such that g_n and g_{n+k} are good Gram points and $g_{n+1}, \dots, g_{n+k-1}$ are bad Gram points.

A Gram block is denoted by the notation $a_1 a_2 \dots a_k$ where k is called the length of the Gram block, and a_i denote the number of roots of $Z(t)$ in the Gram interval $[g_{n+i-1}, g_{n+i})$. So far, no Gram interval has been found with more than 5 zeros, thus the notation is unambiguous. a_1 and a_k must be even while a_2 to a_{k-1} are odd.

Definition 2.4. A Gram block of length k which contains exactly k roots of $Z(t)$ is called regular.

The first and last Gram intervals of a regular Gram block must contain an even number of roots (0 or 2 roots). The internal Gram intervals must all contain an odd number of roots (all of them must contain one root if the end intervals contain 2 and 0 roots. If the end intervals both contain no roots, then one of the internal intervals must contain 3 roots.) Thus, regular Gram blocks must have a pattern of one of the following three forms:

$$\begin{aligned} &21\dots 10, \\ &01\dots 12, \\ &01\dots 131\dots 10 \end{aligned} \tag{2.1}$$

where the notation $1\dots 1$ refers to any string of consecutive 1s, including the zero length string. Following Odlyzko [3] we define three types of Gram blocks. These Gram blocks also have interesting properties for the usual $S(T)$ [4].

Definition 2.5. A Type I Gram Block is a regular Gram Block which has the pattern of zero counts $21\dots 10$. $S(T)$ is 1 within a Type I Gram block.

Definition 2.6. A Type II Gram Block is a regular Gram Block which has the pattern of zero counts $01\dots 12$. $S(T)$ is -1 within a Type II Gram block.

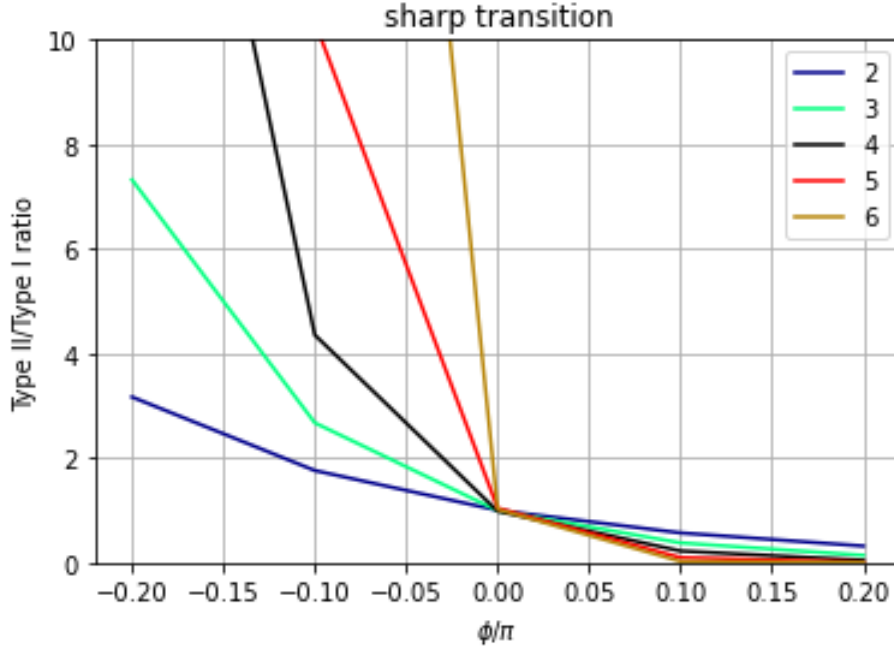


Figure 1: Plot of ratios in Table 1, $t = 10^{12}$.

Definition 2.7. A Type III Gram Block is a regular Gram Block which has the pattern of zero counts 01...131...10. $S(T)$ is -1 at the beginning of a Type III Gram block and transitions to -1 within the block.

A regular Gram block of length 1 is a special case and is not covered by any of the three forms stated above. Note that the sequence of zero counts in a Type II Gram block is the reverse of the sequence of zero counts in a Type I Gram block of the same length. Gram blocks which are not regular have more complex patterns of zero counts on Gram intervals.

3 Conclusions

Tables 1 and 2 show the sharp transition in the *Type II/Type I* Gram block ratio. Figures 1 and 2 show the sharp transition visually. The ratio is greater than one and increases exponentially with the Gram block length when ϕ , the generalized Gram angle, is negative. The ratio is lesser than one and decreases exponentially with the Gram block length when ϕ , the generalized Gram angle, is positive. At the usual Gram points, the ratio is one, thus, there is a forward backward symmetry in the occurrence of Gram block patterns for the

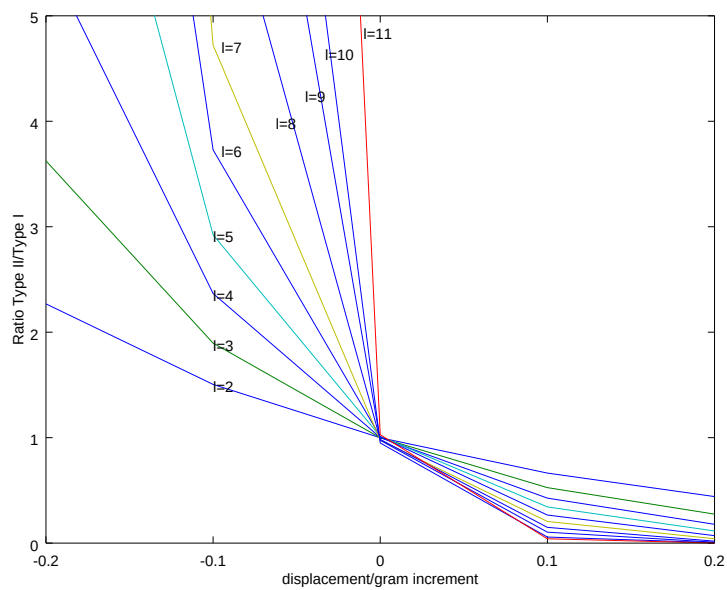


Figure 2: Plot of ratios in Table 2, $t = 10^{28}$.

<i>Length of</i> <i>Gram</i>	<i>Generalized</i>	<i>Gram</i>	<i>angle</i>	ϕ	<i>prod</i>	<i>prod</i>
<i>block</i>	-0.2π	-0.1π	0.0π	0.1π	0.2π	0.1π
2	3.167	1.760	1.000	0.567	0.316	1.000
3	7.323	2.672	0.993	0.376	0.138	1.011
4	20.958	4.352	0.982	0.221	0.043	0.901

Table 1: Sharp transition in *Type II/Type I* Gram block ratio. The ratio is greater than one and increases exponentially with the Gram block length when ϕ , the generalized Gram angle, is negative. The ratio is lesser than one and decreases exponentially with the Gram block length when ϕ , the generalized Gram angle, is positive. Notice that the product of the values at plus and minus ϕ are approximately 1. The statistics are from 1 million Gram intervals at $t = 10^{12}$.

<i>Length of</i> <i>Gram</i>	<i>Generalized</i>	<i>Gram</i>	<i>angle</i>	ϕ	<i>prod</i>	<i>prod</i>
<i>block</i>	-0.2π	-0.1π	0.0π	0.1π	0.2π	0.1π
2	2.269	1.505	0.999	0.664	0.441	1.001
3	3.625	1.896	0.999	0.526	0.275	0.997
4	5.589	2.367	1.001	0.427	0.178	0.995
5	8.850	2.923	1.012	0.343	0.115	1.018
6	14.373	3.729	1.009	0.266	0.071	1.020
7	23.962	4.722	0.975	0.205	0.041	0.982
8	51.791	6.715	0.984	0.149	0.018	0.932
9	116.632	10.130	0.975	0.104	0.009	1.050

Table 2: Sharp transition in *Type II/Type I* Gram block ratio. Notice that the product of the values at plus and minus ϕ are approximately 1. The statistics are from 10 million Gram intervals at $t = 10^{28}$.

usual Gram points. The forward backward symmetry manifests itself in a more complex way [8] for points away from the usual Gram points.

Acknowledgments and Funding Statement

The study was done as an independent researcher. There was no external funding.

Ethical Compliance

No procedures were performed involving human participants in the study.

Data Availability Statement

The computer programs used during the current study are available from the corresponding author on reasonable request.

Conflict of Interest declaration

The authors declare that they have no affiliations with or involvement in any organization or entity with any financial interest in the subject matter or materials discussed in this manuscript.

References

- [1] O. Shanker, “Good to Bad Gram Point Ratio For Riemann Zeta Function”, *Experimental Mathematics* **30**, 76-85, tinyurl.com/mwd5uwc5(2021)
- [2] O. Shanker, “Symmetry properties of distribution of Riemann Zeta Function values on critical axis” report, tinyurl.com/47wj57b3, (2018).
- [3] A. Odlyzko, “The 10^{20} -th Zero of the Riemann Zeta Function and 175 Million of its Neighbors”, report, <http://www.dtc.umn.edu/~odlyzko/unpublished/zeta.10to20.1992.pdf>, (1992)
- [4] H. M. Edwards, “Riemann’s Zeta Function,” Academic Press, (1974)
- [5] O. Shanker, “Neural Network prediction of Riemann zeta zeros” *Advanced Modeling and Optimization*, **14**, 717-728, (2012), tinyurl.com/4scve3nj.
- [6] O. Shanker, “Random Matrix Theory explanation for Riemann Zeta Value Distribution Symmetry” report, <https://tinyurl.com/ywhy4jsy>, (2022).
- [7] O. Shanker, “Generalised Zeta Functions and Self-Similarity of Zero Distributions”, *J. Phys. A* **39**(2006), 13983-13997
- [8] O. Shanker, “Universality of Riemann Zeta Function value distribution on critical axis” report, tinyurl.com/yvbd2je6, (2020).